

Daily Review

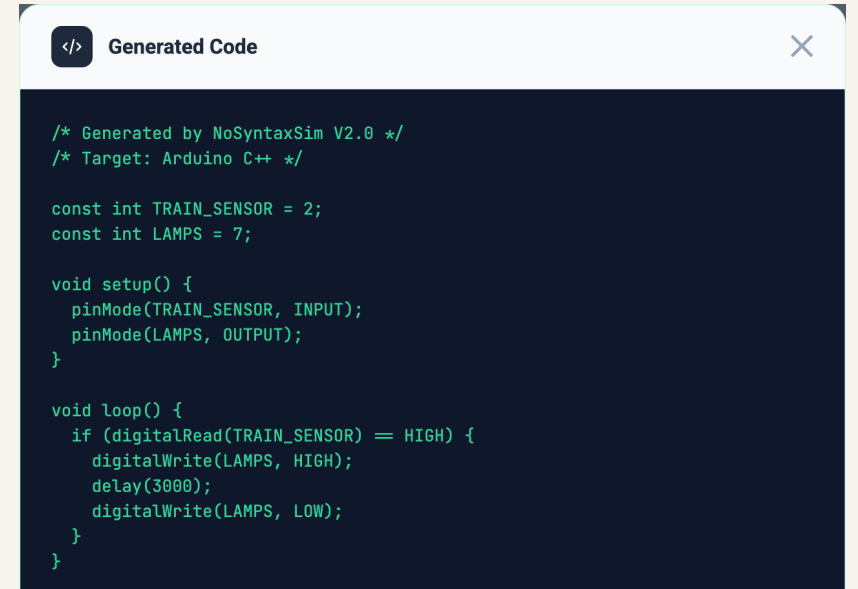


- 1 The diamond is the ... symbol.
- 2 x3? makes what kind of loop?
- 3 Every delay box needs its...
- 4 A loop going up needs an...
- 5 An input becomes which box?

Reading Programs

The exam writes a flowchart as lines.
Can you read them?

National 5 Engineering Science

A screenshot of a code editor window with a white title bar containing a code icon and the text 'Generated Code', and a close button on the right. The editor has a dark blue background and displays C++ code in a light green monospace font. The code is for an Arduino project, defining a sensor pin and several lamps, and implementing a loop that turns the lamps on and off based on a sensor reading.

```
</> Generated Code ×

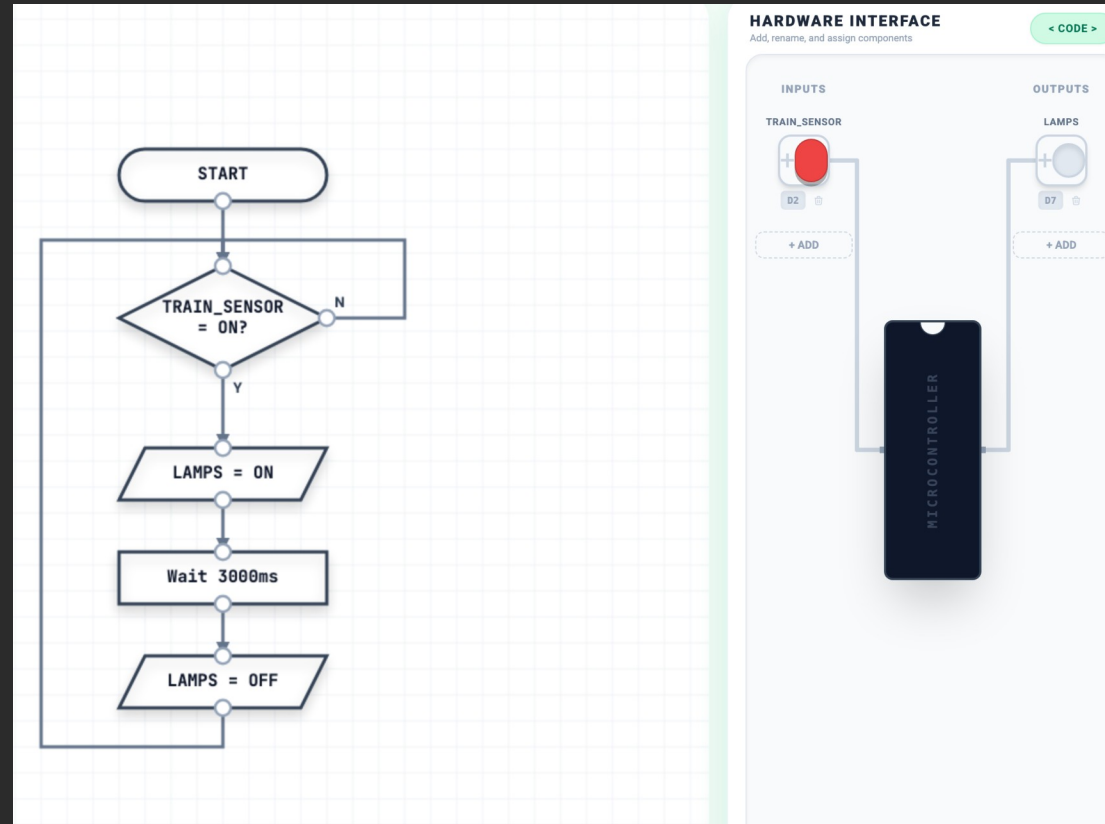
/* Generated by NoSyntaxSim V2.0 */
/* Target: Arduino C++ */

const int TRAIN_SENSOR = 2;
const int LAMPS = 7;

void setup() {
  pinMode(TRAIN_SENSOR, INPUT);
  pinMode(LAMPS, OUTPUT);
}

void loop() {
  if (digitalRead(TRAIN_SENSOR) == HIGH) {
    digitalWrite(LAMPS, HIGH);
    delay(3000);
    digitalWrite(LAMPS, LOW);
  }
}
```

Here is the flowchart. Written as lines, which line is the delay?



A flowchart in NoSyntaxSim

PART 1

Reading the Exam's Program

Line by line, box by box.

By the end of this part...



- 1 I can say what each line of an exam program does
- 2 I can find a line from what it does
- 3 I can say how many times a counter loop runs

The Words the Exam Uses



switch on 7

pin 7 on

pause 500

wait 500 ms = 0.5 s

main: label_1:

a name for a line

goto main

jump to that line

if ... then label_1

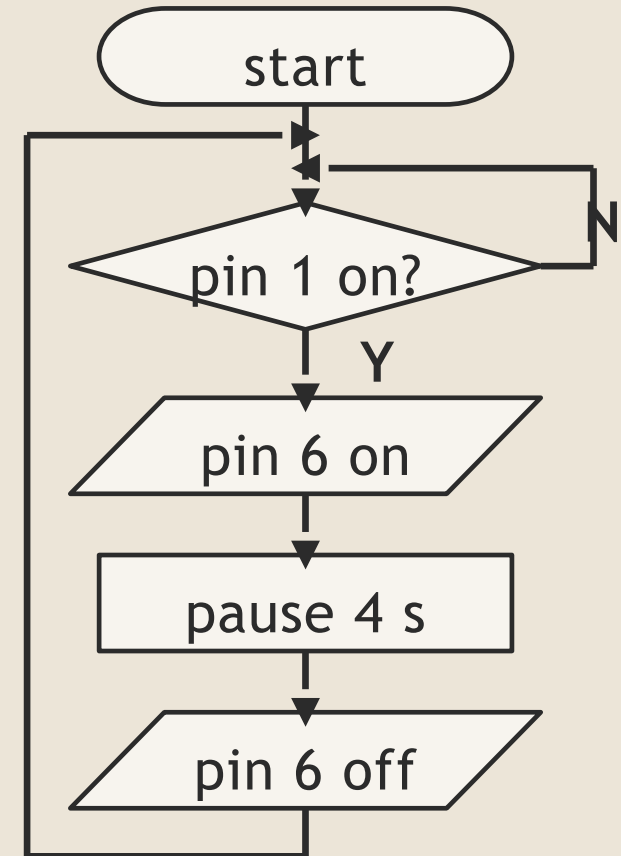
a decision

let count = ...

a counter

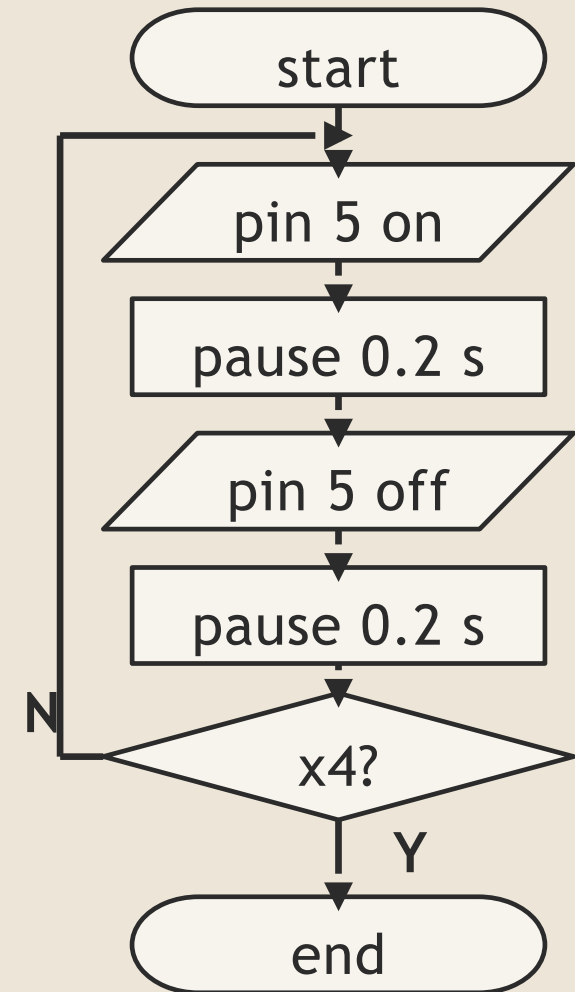
One Program, One Flowchart

```
1  main:    if input1 is off then main  
2           switch on 6  
3           pause 4000  
4           switch off 6  
5           goto main
```



Counting: The Fixed Loop in Code

```
1  main:    let count = 0
2  label_1: switch on 5
3           pause 200
4           switch off 5
5           pause 200
6           let count = count + 1
7           if count = 4 then label_2
8           goto label_1
9  label_2: end
```



Read It



```
1  main:      if input2 is off then main
2              let count = 0
3  label_1:   switch on 7
4              pause 1500
5              switch off 7
6              let count = count + 1
7              if count = 8 then label_2
8              goto label_1
9  label_2:   pause 2500
10             goto main
```

- 1 A line with a delay?
- 2 Times pin 7 flashes?
- 3 Lines 3-8: what loop?
- 4 What does line 5 do?
- 5 How long is line 9?

Model Answer: The Function of a Line (1 mark)



```
1  main:      if input1 is off then main
2              switch on 6
3              pause 4000
4              switch off 6
5              goto main
```

Describe the function of line 5.

Answer (1)

It sends the program back to line 1, so it repeats forever.

Scores 0

“It goes to main.”

Checkpoint



- 1 pause 750 waits how long?
- 2 What does 'goto main' make?
- 3 Why is a label needed?

Plenary: Part 1



- 1 switch off 4 means...
- 2 pause is measured in...
- 3 if count = 5 then label_2 is a...
- 4 let count = count + 1 does what?
- 5 'Goes to main' on its own scores...

Look back — by the end of this part...



- 1 I can say what each line of an exam program does
- 2 I can find a line from what it does
- 3 I can say how many times a counter loop runs

Daily Review



- 1 pause 1500 waits...
- 2 goto label_1 jumps to...
- 3 let count = 0 does what?
- 4 A line with a delay starts with...
- 5 'Goes to main' alone scores...

PART 2

When It Goes Wrong

It should beep 10 times. It never stops. Why?

By the end of this part...



- 1 I can trace a counter loop, pass by pass
- 2 I can explain why a program does not repeat
- 3 I can turn a flowchart into a program

Trace It, Pass by Pass



```
1  main:      let count = 0
2  label_1:   switch on 4
3              let count = count + 1
4              switch off 4
5              if count = 3 then label_2
6              goto label_1
7  label_2:   end
```

Run count = 3? line

Model Answer: Why It Does Not Repeat (2 marks)



```
1  main:      let count = 0
2  label_1:   switch on 7
3              pause 300
4              switch off 7
5              pause 300
6              let count = count + 1
7              if count = 10 then label_2
8              goto main
9  label_2:   switch on 5
10             end
```

Explain why lines 2-7 don't run 10 times.

Cause (1)

Line 8 goes back to line 1,

Effect (1)

so the count resets: never 10.

Watch Out: Loop Back Below the Count

THE MISTAKE

- 8 goto main
- The count goes back to 0.
- It never reaches 10.

THE CORRECT ANSWER

- 8 goto label_1
- The count keeps going up.
- It stops at 10.

Spot the Mistake



```
1  main:      let count = 0
2  label_1:   switch on 6
3             pause 500
4             switch off 6
5             pause 500
6             if count = 3 then label_2
7             goto label_1
8  label_2:   switch on 7
9             end
```

It should

flash pin 6 three
times, then pin 7 on

It does

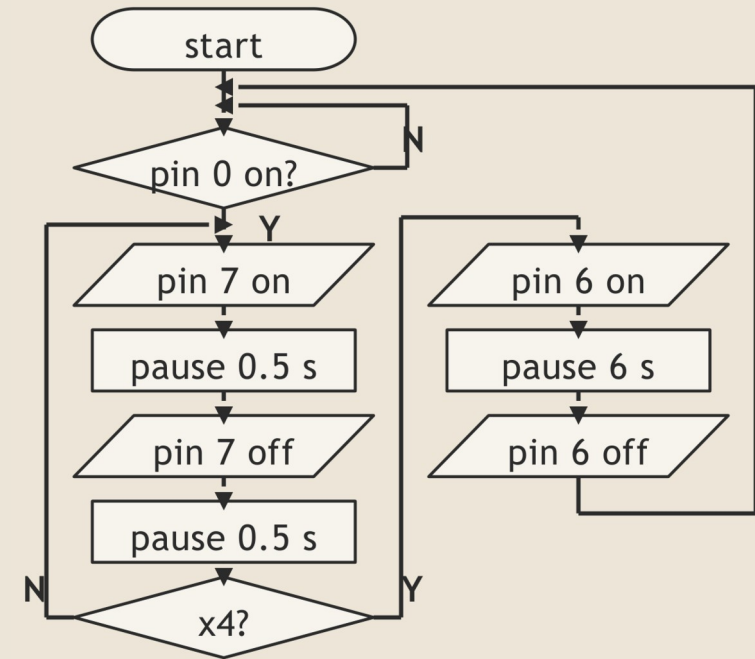
flash pin 6 for ever

Your job

find the missing line

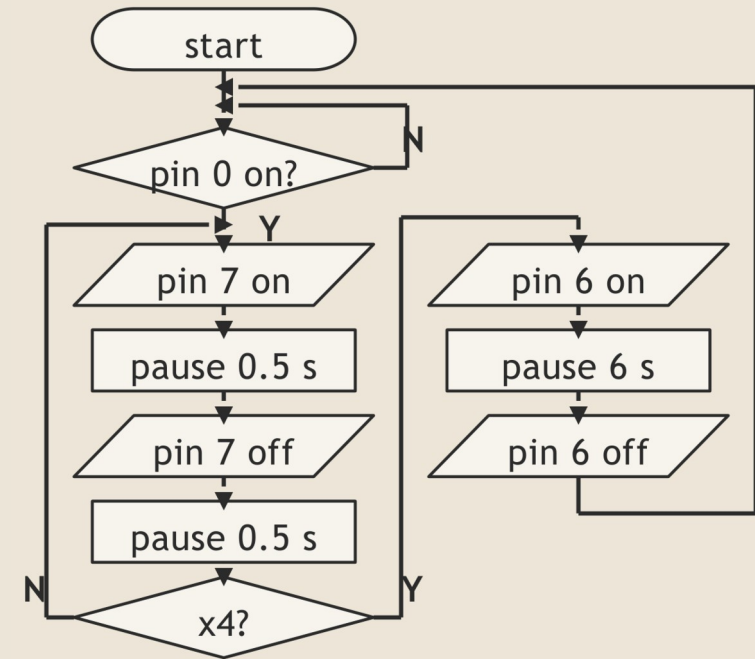
Flowchart to Program (1 of 2)

```
1  main:      if input0 is off then main
2              let count = 0
3  label_1:   switch on 7
4              pause 500
5              switch off 7
6              pause 500
7              let count = count + 1
8              if count = 4 then label_2
9              goto label_1
```



Flowchart to Program (2 of 2)

```
10 label_2:  switch on 6
11           pause 6000
12           switch off 6
13           goto main
```



Your Turn: Program to Flowchart



```
1  main:      if input3 is off then main
2              switch on 5
3              switch on 4
4              pause 2000
5              switch off 5
6              pause 1000
7              switch off 4
8              goto main
```

Draw

the flowchart for this program

Check

every pin, every unit,
every arrow

Your Turn: The Answer



start

Past Papers: Reading Programs



A microcontroller based system is used to detect the bottles.

The program used to count six bottles is shown below.

line	program
1	main: let count = 0
2	check: if input2 is off then check
3	let count = count + 1
4	if count = 6 then label_1
5	goto check
6	label_1: switch on 7
7	pause 500
8	switch off 7
9	goto main

(c) State the line number that contains a time delay.

2023 Q9(c)

2. A microcontroller program is shown.

line	program
1	main: if input1 is on then label_1
2	goto main
3	label_1: let count = 0
4	label_2: switch on pin 7
5	pause 500
6	switch off pin 7
7	pause 500
8	let count = count + 1
9	if count = 15 then label_3
10	goto label_2
11	label_3: end

(a) State a line number that contains a time delay.

Line number _____

(b) State the number of times that pin 7 will turn on and then off.

(c) State the name of the type of loop used to control the number of times pin 7 turns on and off.

2025 Q2

A program used in a different control system is listed below.

line	program
1	main: let count = 0
2	label_1: switch on 4
3	switch on 5
4	pause 600
5	switch off 4
6	switch off 5
7	pause 600
8	let count = count + 1
9	if count = 20 then label_2
10	goto main
11	label_2: if Input0 is on then label_3
12	goto label_2
13	label_3: switch on 7
14	pause 3000
15	switch off 7
16	goto main

(b) Describe the function of line 16 in the program.

Lines 2 to 9 should repeat twenty times before moving on to line 11.

During testing an electronic engineer found that this did not happen.

(c) Explain why lines 2 to 9 did not repeat twenty times before moving on to line 11.

2022 Q9(b)(c)

Past Papers: Answers



1 2023 Q9(c)

2 2025 Q2(a), (b)

3 2025 Q2(c)

4 2022 Q9(b)

5 2022 Q9(c) (2)

Past Papers: Find Them All



Past papers

Year	Question
2022	Q9(b), (c)
2023	Q9(c)
2025	Q2



By topic

Checkpoint



- 1 A count never reaches its limit. Look for?
- 2 No count + 1 line: what happens?
- 3 Why must the loop go to label_1?

Plenary: Part 2



- 1 Explain questions need a cause and...
- 2 The words that join them?
- 3 x4? becomes how many lines?
- 4 The arrow back up becomes...
- 5 The N loop of line 1 goes to...

Look back — by the end of this part...



- 1 I can trace a counter loop, pass by pass
- 2 I can explain why a program does not repeat
- 3 I can turn a flowchart into a program

Daily Review



- 1 pause 2000 waits...
- 2 A goto above the count reset makes...
- 3 Explain questions need...
- 4 switch on 3 means...
- 5 The flowchart's x3? becomes...

PART 3

Two More Ways to Write It

The booklet's pseudocode, then real Arduino C.

By the end of this part...



- 1 I can describe what `setup()` and `loop()` do
- 2 I can match a line of Arduino C to what it does
- 3 I can write pseudocode for a simple program

Three Ways to Write One Program

The exam

```
main:  
switch on 5  
pause 500  
switch off 5  
pause 1000  
goto main
```

The booklet

```
loop  
high buzzer  
pause 0.5 seconds  
low buzzer  
pause 1 second  
end loop
```

Arduino C

```
void loop() {  
  digitalWrite(5, HIGH);  
  delay(500);  
  digitalWrite(5, LOW);  
  delay(1000);  
}
```

The Standard Layout



setup()

runs once: set each pin as input or output

loop()

runs for ever: the control program

```
void setup() {  
    pinMode(5, OUTPUT);  
}
```

```
void loop() {  
    // the program  
}
```

The Commands You Need



pinMode(4, OUTPUT);

pin 4 is an output (in setup)

digitalWrite(4, HIGH);

pin 4 on • LOW: off

delay(250);

wait 250 ms

digitalRead(2)

read the input on pin 2

Let the Simulator Write It

Flowchart in

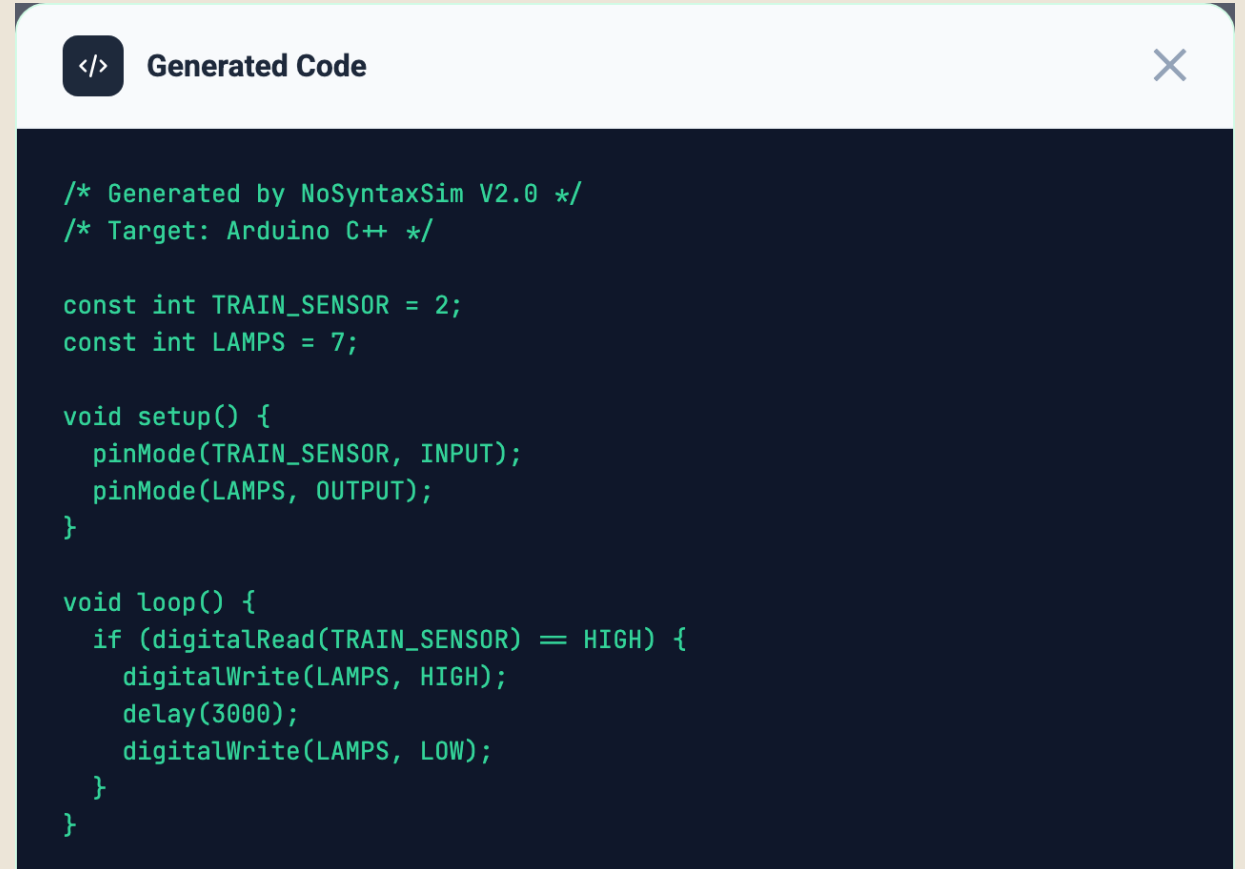
Arduino C out

The decision

if (digitalRead(...) ==
HIGH)

The wait

delay(3000)



```
</> Generated Code X

/* Generated by NoSyntaxSim V2.0 */
/* Target: Arduino C++ */

const int TRAIN_SENSOR = 2;
const int LAMPS = 7;

void setup() {
  pinMode(TRAIN_SENSOR, INPUT);
  pinMode(LAMPS, OUTPUT);
}

void loop() {
  if (digitalRead(TRAIN_SENSOR) == HIGH) {
    digitalWrite(LAMPS, HIGH);
    delay(3000);
    digitalWrite(LAMPS, LOW);
  }
}
```

What Does It Do?



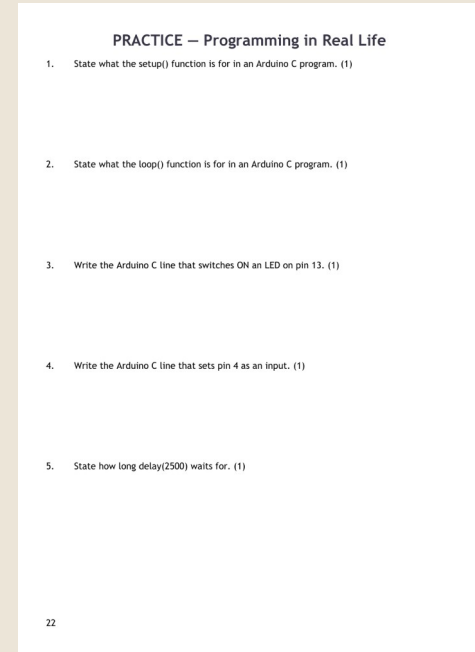
- 1 `pinMode(3, INPUT);`
- 2 `digitalWrite(7, HIGH);`
- 3 `delay(250);`
- 4 `digitalWrite(7, LOW);`
- 5 `digitalRead(2)`

PRACTICE: Programming in Real Life



Booklet pages 22-23

Q1 to Q6, on your own.
Q7 and Q8 are for Lesson 4.



12 min



Programming in Real Life: Answers



1 Q1 setup()

2 Q2 loop()

3 Q3 and Q4

4 Q5 delay(2500)

5 Q6 (4 marks)

Plenary: Part 3



- 1 `setup()` runs how often?
- 2 `loop()` runs...
- 3 `delay(1000)` waits...
- 4 The booklet's pause 1 second in the exam?
- 5 Which tool writes Arduino C for you?

Extension

EXTRA PRACTICE



- 1 Rewrite the beeper with a goto bug. Effect?
- 2 `for (int i = 0; i < 5; i++)` runs how often?
- 3 Why does Arduino C need no 'goto main'?

Look back — by the end of this part...



- 1 I can describe what `setup()` and `loop()` do
- 2 I can match a line of Arduino C to what it does
- 3 I can write pseudocode for a simple program

LOOKING AHEAD

Lesson 4: Build, Test and Fix

Put it on the Uno. Then find what is wrong with it.

Course Specification

Digital electronic control systems

- using suitable commands (high, low, for...next, if...then, pause, end (or their equivalents)) to design programs to solve simple control problems, involving time delays and continuous and fixed loops

Credits (1 of 2)

Title slide (NoSyntaxSim): NoSyntaxSim V2 screenshot (Generate Code). Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

The big question (NoSyntaxSim): NoSyntaxSim V2 screenshot. Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

Model Answer: The Function of a Line (2022 Q9(b) MI): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2022 Q9(b)(c), 2023 Q9(c), 2025 Q2.

Model Answer: Why It Does Not Repeat (2022 Q9(c) MI): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2022 Q9(b)(c), 2023 Q9(c), 2025 Q2.

Watch Out (2022 Q9(c) MI): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2022 Q9(b)(c), 2023 Q9(c), 2025 Q2.

Credits (2 of 2)

Past Papers: Reading Programs: © Qualifications Scotland (SQA). N5 Engineering Science 2022 Q9(b)(c), 2023 Q9(c) and 2025 Q2, question papers and marking instructions.

Course Specification: © Qualifications Scotland (SQA). Course spec v2.0, p.5

Let the Simulator Write It: NoSyntaxSim V2 screenshot (Generate Code). Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

PRACTICE: Programming in Real Life (booklet page): Booklet 4b pages by R Stewart.

R Stewart - October 2026